

Muhammad Umer Khan Niazi

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EDUCATION

University of Engineering and Technology

BS Artificial Intelligence

Lahore, Pakistan

2025 – Present

Forman Christian College

Intermediate Computer Science

Lahore, Pakistan

2022 – 2024

PROJECTS

News NLP Pipeline | *Python, BERTopic, scikit-learn, spaCy, Transformers, Streamlit*

2026 – Present

- Extracted and processed 350k+ headlines from Dawn News (2010–2025) using an end-to-end NLP pipeline
- Implemented BERTopic with UMAP and K-Means for unsupervised topic discovery across 15 years of articles without manual labeling
- Developed downstream ML models for sentiment analysis, named entity recognition (NER), and category classification
- Deployed an interactive Streamlit dashboard to visualize topic distributions and temporal coverage trends

WMUS | *Python, curses*

2026 – Present

- Developed a keyboard-centric terminal music player for Windows to enable lightweight, seamless audio playback
- Architected a modular system separating audio playback, UI rendering, and state management
- Implemented fuzzy search, vim-style navigation, and customizable keybindings
- Engineered a metadata caching layer to eliminate redundant library scans and distributed the tool via a versioned installer

Himalayan Wildlife Classifier | *Python, PyTorch, Streamlit*

2026

- Fine-tuned a custom classification head on a frozen ResNet18 backbone to identify three endangered Himalayan species, achieving 98.16% validation accuracy
- Applied aggressive data augmentation techniques to mitigate overfitting on a limited dataset
- Integrated an 80% confidence threshold to flag uncertain predictions and prevent forced misclassifications
- Evaluated the model against out-of-distribution inputs, documenting failure cases caused by shared backgrounds and taxonomic similarities

Waddle Away | *Godot, GDScript*

2026

- Built and shipped a small Godot game with custom player movement, obstacle spawning, and gameplay systems
- Implemented game states, collision handling, and progression-based difficulty adjustments

TECHNICAL SKILLS

Languages: Python, C++, GDScript

Tools & Platforms: Git, Linux, Streamlit, SQLite, Godot

Machine Learning & NLP: NumPy, Pandas, Matplotlib, scikit-learn, Seaborn, PyTorch, Transformers, Hugging Face, BERTopic